Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 7 — Integration
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第7章 · 积分
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1.

For $x > 0$,

(a) find $\int \left(3e^{2x} - \frac{4}{x}\right) dx$, (4)

(b) hence evaluate $\int_1^2 \left(3e^{2x} - \frac{4}{x}\right) dx$ exactly. (3)

(Total for Question 1 is 7 marks)

2. P3-C07-R02 · P3 · Chapter 7

The curve C has gradient $dy/dx = 6\cos 3x - 2\sin x$ and passes through $(0, 5)$.

(a) Find the equation of C. (3)

(b) Find the exact value of y at $x = \pi/6$. (2)

(Total for Question 2 is 5 marks)

For $x > -1/2$,

(a) find $\displaystyle\int \frac{6}{2x+1}\,dx$, (4)

(b) evaluate $\displaystyle\int_0^1 \frac{6}{2x+1}\,dx$ exactly. (2)

(Total for Question 3 is 6 marks)

4.

P3-C07-R04 · P3 · Chapter 7

For real x, let $F(x) = (x^2 + 1)^4$.

(a) Find $F'(x)$. (2)

(b) Hence evaluate $\displaystyle\int_0^1 x(x^2 + 1)^3 \, dx$ exactly. (3)

(Total for Question 4 is 5 marks)

R is bounded by $y = e^x$, the x-axis and the lines $x = 0, x = \ln 3$.

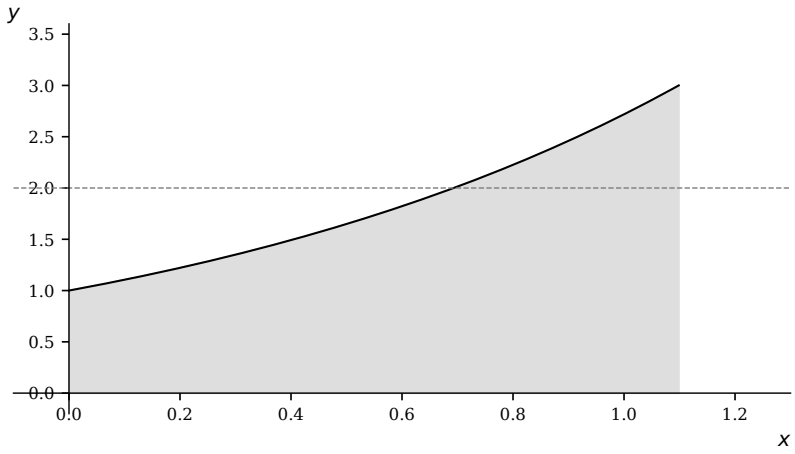


Figure 5

(a) Find its exact area. (3)

(b) The horizontal line $y = 2$ is drawn.

(i) find its intersection with $y = e^x$, (2)

(ii) find the exact area of the part of R above $y = 2$. (3)

(Total for Question 5 is 8 marks)

Find each indefinite integral.

(a) $\int \cos(5x - 2)\, dx.$ (4)

(b) $\int \sec^2(3x)\, dx.$ (4)

(Total for Question 6 is 8 marks)

For $x \geq 0$, $f(x) = \ln(x^2 + 4)$.

(a) Find $f'(x)$. (2)

(b) Hence find $\displaystyle\int_0^2 \frac{3x}{x^2 + 4}\,dx$ exactly. (3)

(Total for Question 7 is 5 marks)

The curve $C : y = \sin x$ on $[0, \pi]$ is cut by $y = 1/2$.

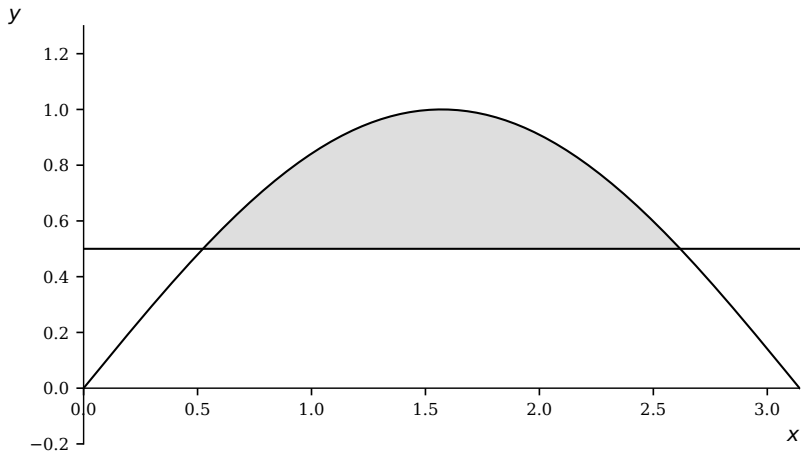


Figure 8

(a) Find the two intersection points. (2)

(b) Find the exact area of the region enclosed by the curve and the line $y = 1/2$. (4)

(Total for Question 8 is 6 marks)

For $x > 0$,

(a) find $\int \left(\frac{x^2 + 3x + 2}{x} \right) dx$, (4)

(b) evaluate this integral between $x = 1$ and $x = 3$. (3)

(Total for Question 9 is 7 marks)

$F(x) = e^{x^2}$.

(a) Find $F'(x)$. (2)

(b) Hence find $\int_0^{\sqrt{\ln 3}} 4xe^{x^2}\,dx$ exactly. (3)

(Total for Question 10 is 5 marks)

For $0 < x < \pi$,

(a) show that $\dfrac{d}{dx}\ln(\sin x) = \cot x$, (3)

(b) hence evaluate $\displaystyle\int_{\pi/6}^{\pi/2} \cot x \, dx$. (3)

(Total for Question 11 is 6 marks)

12. P3-C07-R12 · P3 · Chapter 7

The curve C satisfies $dy/dx = 4e^{-2x} + 3$ and passes through $(0, 1)$.

(a) Find the equation of C. (3)

(b) Find the exact y-coordinate when $x = \ln 2$. (3)

(Total for Question 12 is 6 marks)

Use double-angle formulae to evaluate exactly:

(a) $\displaystyle\int_0^{\pi/2} \sin^2 x \, dx,$ (4)

(b) $\displaystyle\int_0^{\pi/4} \cos^2 x \, dx.$ (4)

(Total for Question 13 is 8 marks)

Let $F(x) = \sin^3 x$.

(a) Find $F'(x)$. (2)

(b) Hence evaluate $\displaystyle\int_0^{\pi/2} \sin^2 x \cos x \, dx$. (3)

(Total for Question 14 is 5 marks)

The line $y = 3$ meets $C : y = e^{2x} + 1$ at P. Region R lies between the line, curve and the y-axis.

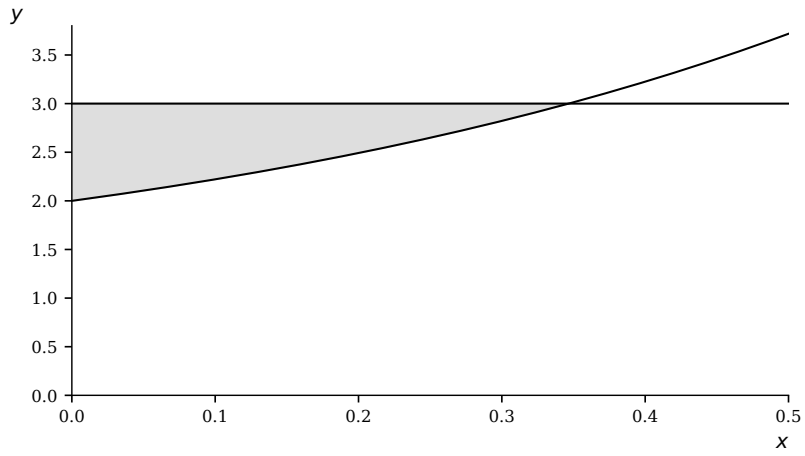


Figure 15

(a) Find P exactly. (3)

(b) Find the exact area of R. (4)

(Total for Question 15 is 7 marks)

For $x > -1$,

(a) find $\int \left(\frac{2}{x+1} + 3(x+1)^2 \right) dx$, (4)

(b) evaluate the integral from 0 to 1 exactly. (2)

(Total for Question 16 is 6 marks)

$F(x) = \dfrac{1}{2x+3}$ for $x > -3/2$.

(a) Find $F'(x)$. (2)

(b) Hence evaluate $\displaystyle\int_0^1 \frac{5}{(2x+3)^2}\,dx$. (3)

(Total for Question 17 is 5 marks)

The gradient of a curve is $dy/dx = 2\sec^2 x - \sin x$ on $-\pi/2 < x < \pi/2$. The curve passes through $(0, 4)$.

(a) Find its equation. (3)

(b) Find the exact value of y at $x = \pi/3$. (3)

(Total for Question 18 is 6 marks)

For real x, $F(x) = \ln(1 + e^{2x})$.

(a) Find $F'(x)$. (2)

(b) Hence evaluate $\displaystyle\int_0^{\ln 2} \frac{e^{2x}}{1 + e^{2x}}\,dx$ exactly. (3)

(Total for Question 19 is 5 marks)

The graph of $f(x) = \cos x - \sin x$ on $[0, \pi/2]$ crosses the x-axis once.

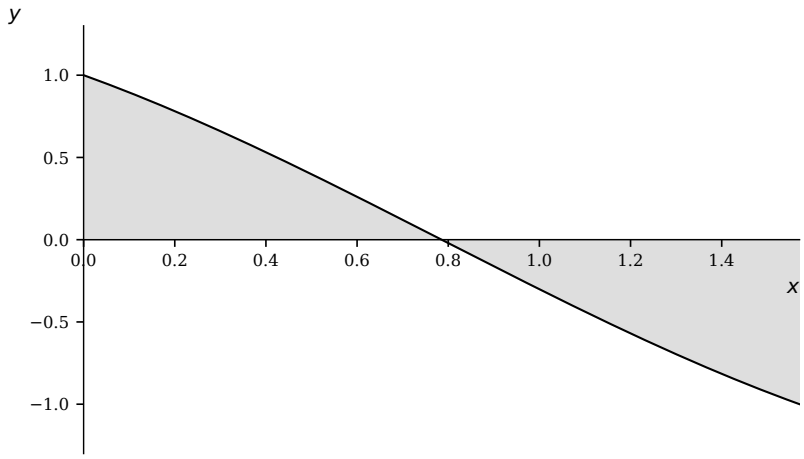


Figure 20

(a) Find the zero exactly. (2)

(b) Find the total positive area between the curve and the x-axis. (4)

(Total for Question 20 is 6 marks)

21.

For $x > 0$, $F(x) = x^2 \ln x - \frac{1}{2}x^2$.

(a) Show that $F'(x) = 2x \ln x$. (3)

(b) Hence evaluate $\displaystyle\int_1^e x \ln x \, dx$. (3)

(Total for Question 21 is 6 marks)

A rate is modelled by $R(t) = 6e^{-0.5t}$ units per minute for $t \geq 0$. The total amount accumulated by time t is defined as $\int_0^t R(s)\,ds$.

(a) Find the exact accumulated amount at time t. (3)

(b) Find the time when 9 units have accumulated. (3)

(Total for Question 22 is 6 marks)

Use the identity $\cos 2x = 2\cos^2 x - 1$.

(a) Find $\displaystyle\int (4\cos^2 x - 1)\,dx$. (4)

(b) Find the exact integral from 0 to $\pi/3$. (2)

(Total for Question 23 is 6 marks)

$f(x) = (3x + 2)^4$.

(a) Find an antiderivative of f. (3)

(b) Find $a > 0$ such that $\displaystyle\int_0^a f(x)\,dx = 992/15$. (4)

(Total for Question 24 is 7 marks)

Let $F(x) = \cos(\ln x)$ for $x > 0$.

(a) Find $F'(x)$. (2)

(b) Hence evaluate $\displaystyle\int_1^{e^{\pi/2}} \frac{\sin(\ln x)}{x}\,dx$. (3)

(Total for Question 25 is 5 marks)

The curve $C : y = e^{x} - e^{-x}$ is considered on $[-\ln 2, \ln 2]$.

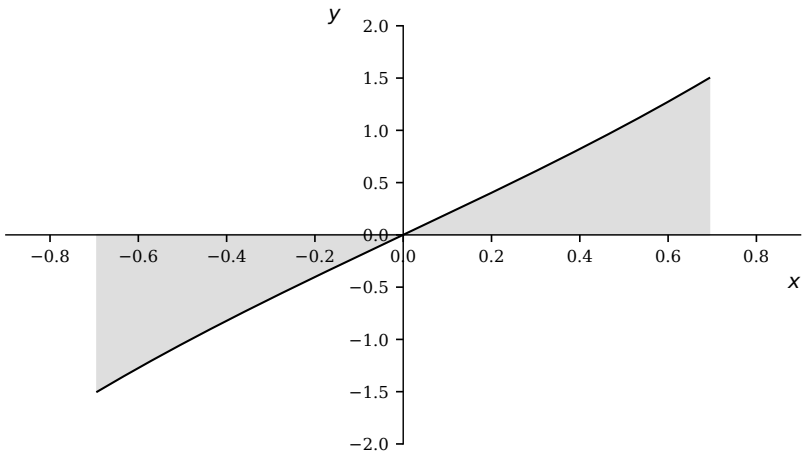


Figure 26

(a) Find its zero and show that its signed integral on this interval is zero. (4)

(b) Find the total area between the curve and the axis on this interval. (4)

(Total for Question 26 is 8 marks)

27. P3-C07-R27 · P3 · Chapter 7

The curve has gradient $dy/dx = 5/(5x + 1)$ for $x > -1/5$, and passes through $(0, 2)$.

(a) Find its equation. (3)

(b) Find where this curve crosses the x-axis. (3)

(Total for Question 27 is 6 marks)

For $0 < x < \pi/2$, $F(x) = \ln(\tan x)$.

(a) Show that $F'(x) = 1/(\sin x \cos x)$. (3)

(b) Hence evaluate $\displaystyle\int_{\pi/6}^{\pi/4} \frac{1}{\sin x \cos x}\, dx$. (3)

(Total for Question 28 is 6 marks)

A function F is given by $F(x) = (x - 2)e^{2x}$.

(a) Show that $F\prime(x) = (2x - 3)e^{2x}$. (2)

(b) Hence, using your answer to (a),

(i) find $\int_0^1 (2x - 3)e^{2x}dx$ exactly. (2)

(ii) find the positive area bounded by $y = (2x - 3)e^{2x}$, the x-axis and the lines $x = 0, x = 1$. (2)

(Total for Question 29 is 6 marks)

For $0 \leq x \leq \pi/4$, the curve C has equation $y = \sec^2 x + 2\cos 2x$.

(a) Find the exact area between C, the x-axis and the lines $x = 0, x = \pi/4$. (3)

(b) The line $y = k$ is drawn across the same vertical strip.

(i) Find k if the signed integral $\int_0^{\pi/4}(y-k)dx$ is zero. (2)

(ii) Explain why this does not mean that the curve is the horizontal line $y = k$. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C07-R01 · 7 marks

(a) · 4 marks

$\frac{3}{2}e^{2x} - 4\ln x + C.$

指数积分除内层常数2；1/x积分为lnx，原域x正。

M1	指数积分
M1	对数积分
A1	两项
B1	常数

(b) · 3 marks

$\frac{3}{2}(e^4 - e^2) - 4\ln 2.$

使用(a)原函数在上限减下限，ln1=0。

M1	正确代上下限
M1	对数化简
A1	精确值

Question 2

P3-C07-R02 · 5 marks

(a) · 3 marks

$y = 2\sin 3x + 2\cos x + 3.$

积分cos3x除3，积分−2sinx为2cosx；点条件给常数3。

M1	三角积分及内层因子
M1	点条件
A1	常数与曲线

(b) · 2 marks

$5 + \sqrt{3}.$

2sinπ/2=2，2cosπ/6=√3，另加3。

M1	正确输入精确值
A1	y

Question 3

P3-C07-R03 · 6 marks

(a) · 4 marks

$3\ln(2x+1)+C$.

对数真数导数2，外系数需为6/2=3，域内真数正。

M1	识别对数导数
M1	内层因子调整
A1	原函数
B1	常数

(b) · 2 marks

$3\ln 3$.

代上下限$3\ln 3-3\ln 1$。

M1	正确上下限
A1	值

Question 4

P3-C07-R04 · 5 marks

(a) · 2 marks

$8x(x^2+1)^3$.

外幂4乘内导2x。

M1	链式
A1	导式

(b) · 3 marks

$15/8$.

被积式为F′/8，原函数F/8，上下限(16−1)/8。

M1	导数识别及比例
M1	上下限
A1	精确值

Question 5

P3-C07-R05 · 8 marks

(a) · 3 marks

2.

e^x恒正，积分[e^x]0至ln3=3−1。

M1	正确积分与界
M1	原函数代值
A1	面积

(b) The horizontal line $y = 2$ is drawn.

(i) · 2 marks

$(\ln 2, 2)$.

e^x2，取ln。

M1	对数反解
A1	点

(ii) · 3 marks

$1 - 2\ln(3/2)$.

只在ln2到ln3，积分(e^x−2)，结果(3−2ln3)−(2−2ln2)。

M1	正确上下界和上减下
M1	积分
A1	精确面积

Question 6

P3-C07-R06 · 8 marks

(a) · 4 marks

$\frac{1}{5}\sin(5x-2)+C$.

正弦导数给余弦，再除输入导数5。

M1	标准三角积分
M1	系数1/5
A1	原函数
B1	常数

(b) · 4 marks

$\frac{1}{3}\tan 3x+C$.

tan3x导数3sec²3x，因此积分除3，取同一连续定义区间。

M1	识别正切导数
M1	输入因子
A1	原函数
B1	常数

Question 7

P3-C07-R07 · 5 marks

(a) · 2 marks

$2x/(x^2+4)$.

对数链式，分子为内部二次式导数。

M1	对数链式
A1	导式

(b) · 3 marks

$\frac{3}{2}\ln 2$.

原函数$(3/2)\ln(x^2+4)$，上下限给$(3/2)\ln(8/4)$。

M1	正确识别比例
M1	上下限及对数商
A1	值

Question 8

P3-C07-R08 · 6 marks

(a) · 2 marks

$(\pi/6, 1/2), (5\pi/6, 1/2).$

sin在半圈取1/2于参考角π/6及补角。

M1	合法角范围
A1	两点

(b) · 4 marks

$\sqrt{3} - \pi/3.$

积分π/6至5π/6的(sinx−1/2)，原函数−cosx−x/2。

M1	正确界和差
M1	原函数
M1	精确代值
A1	面积

Question 9

P3-C07-R09 · 7 marks

(a) · 4 marks

$x^2/2 + 3x + 2\ln x + C$.

先逐项除x得x+3+2/x，最后一项用对数积分。

M1	正确化简
M1	混合积分
A1	三项
B1	常数

(b) · 3 marks

$10 + 2\ln 3$.

多项式部分差(9−1)/2+3(3−1)=10，另加2ln3。

M1	上下限
M1	对数项
A1	值

Question 10

P3-C07-R10 · 5 marks

(a) · 2 marks

$2xe^{x^2}$.

指数链式内导2x。

M1	链式
A1	导式

(b) · 3 marks

4.

原函数2e^{x^2}，上端x^2ln3给6，下端给2。

M1	识别比例
M1	正确根式上限
A1	精确值

Question 11

P3-C07-R11 · 6 marks

(a) · 3 marks

Derivative $\cos x / \sin x = \cot x$.

域内sin正，ln合法；链式化为余切。

M1	对数链式
M1	倒数比化简
A1	目标式

(b) · 3 marks

$\ln 2$.

[lnsinx]上下限=ln1−ln(1/2)。

M1	识别原函数
M1	精确三角值代限
A1	值

Question 12

P3-C07-R12 · 6 marks

(a) · 3 marks

$y = -2e^{-2x} + 3x + 3$.

指数积分除−2；初值−2+C=1给C3。

M1	负指数积分
M1	积分常数条件
A1	方程

(b) · 3 marks

$5/2 + 3\ln 2$.

e^{-2ln2}=1/4，故−1/2+3ln2+3。

M1	指数对数还原
M1	代完整曲线
A1	值

Question 13

P3-C07-R13 · 8 marks

(a) · 4 marks

$\pi/4$.

$\sin^2=(1-\cos2x)/2$，原函数$x/2-\sin2x/4$。

M1	正确半角平方重写
M1	标准积分
M1	上下限
A1	值

(b) · 4 marks

$\pi/8+1/4$.

$\cos^2=(1+\cos2x)/2$，原函数$x/2+\sin2x/4$。

M1	双角重写
M1	积分
M1	精确代限
A1	值

Question 14

P3-C07-R14 · 5 marks

(a) · 2 marks

$3\sin^2 x\cos x$.

三次外幂乘sin的导数cos。

M1	链式
A1	导式

(b) · 3 marks

$1/3$.

原函数$\sin^3 x/3$，上端1下端0。

M1	识别导数比例
M1	上下限
A1	值

Question 15

P3-C07-R15 · 7 marks

(a) · 3 marks

$P = (\frac{1}{2}\ln 2, 3)$.

e^{2x}=2，取ln后除2。

M1	指数方程
M1	自然对数
A1	点

(b) · 4 marks

$\ln 2 - 1/2$.

在0至ln2/2，直线在上，积分2−e^{2x}，原函数2x−e^{2x}/2。

M1	正确差和界
M1	指数积分因子
M1	上下限
A1	面积

Question 16

P3-C07-R16 · 6 marks

(a) · 4 marks

$2\ln(x+1)+(x+1)^3+C$.

两项分别识别ln与三次幂导数；输入导数1。

M1	对数积分
M1	幂识别
A1	两项
B1	常数

(b) · 2 marks

$7+2\ln 2$.

原函数差：2ln2+8−1。

M1	正确代限
A1	精确值

Question 17

P3-C07-R17 · 5 marks

(a) · 2 marks

$-2/(2x+3)^2$.

写负一次幂，乘内导2。

M1	负幂链式
A1	导式

(b) · 3 marks

$1/3$.

原函数−5/[2(2x+3)]，上下限−1/2+5/6=1/3。

M1	识别比例与负号
M1	上下限
A1	值

Question 18

P3-C07-R18 · 6 marks

(a) · 3 marks

$y = 2\tan x + \cos x + 3$.

积分$\sec^2$得tan，−sin得cos，初值给C3。

M1	两项标准积分
M1	初值
A1	曲线

(b) · 3 marks

$7/2 + 2\sqrt{3}$.

域内π/3合法；tan √3、cos1/2，另加3。

M1	精确三角值
M1	完整式代入
A1	y

Question 19

P3-C07-R19 · 5 marks

(a) · 2 marks

$2e^{2x}/(1+e^{2x})$.

对数链式内导2e^{2x}。

M1	链式
A1	导式

(b) · 3 marks

$\frac{1}{2}\ln(5/2)$.

原函数F/2；上端e^{2ln2}=4，差半ln5−半ln2。

M1	识别半倍原函数
M1	指数代限
A1	对数精确值

Question 20

P3-C07-R20 · 6 marks

(a) · 2 marks

$x = \pi/4$.

cos=sin，两者均非负，锐域唯一tan1。

M1	正确方程与域
A1	零点

(b) · 4 marks

$2(\sqrt{2}-1)$.

在0至π/4积分f，在π/4至π/2积分−f，原函数sinx+cosx；两块均√2−1。

M1	在零点分段
M1	正负面积正确
M1	原函数代限
A1	总面积

Question 21

P3-C07-R21 · 6 marks

(a) · 3 marks

Product derivative $2x\ln x + x - x = 2x\ln x$.

乘积法则产生的x与后项导数x抵消。

M1	乘积求导
M1	正确抵消
A1	导式

(b) · 3 marks

$(e^2+1)/4$.

使用题目已给原函数的一半；F(e)=$e^2/2$，F1=$-1/2$。

M1	识别一半F
M1	两限值
A1	精确值

Question 22

P3-C07-R22 · 6 marks

(a) · 3 marks

$12(1-e^{-t/2})$.

原函数−12e^{-s/2}，代t和0。

M1	指数积分含负因子
M1	上下限
A1	累积式

(b) · 3 marks

$t=2\ln 4$.

12(1−e^{-t/2})=9给指数1/4，取ln求时间。

M1	累积阈值方程
M1	对数解
A1	精确时间

Question 23

P3-C07-R23 · 6 marks

(a) · 4 marks

$x + \sin 2x + C$.

被积式=1+2cos2x，积分为x+sin2x。

M1	正确双角重写
M1	标准积分
A1	两项
B1	常数

(b) · 2 marks

$\pi/3 + \sqrt{3}/2$.

代原函数，sin2π/3=√3/2。

M1	精确代限
A1	值

Question 24

P3-C07-R24 · 7 marks

(a) · 3 marks

$(3x+2)^5/15$.

五次幂求导带15倍，所以除15。

M1	幂加1
M1	内层因子3
A1	原函数

(b) · 4 marks

$a=2/3$.

[(3a+2)^5−32]/15=992/15给(3a+2)^5=1024=4^5，得3a+2=4，所以a=2/3。

M1	正确定积分方程
M1	五次正根
M1	正域筛选
A1	a

Question 25

P3-C07-R25 · 5 marks

(a) · 2 marks

$-\sin(\ln x)/x$.

外余弦导数负正弦，内ln导数1/x。

M1	复合链式
A1	导式

(b) · 3 marks

1.

原函数−cosln⁡x，上端−cosπ/2=0，下端−cos0=−1。

M1	识别负F
M1	对数上限
A1	值

Question 26

P3-C07-R26 · 8 marks

(a) · 4 marks

Zero $x = 0$; antiderivative $e^{x} + e^{-x}$ has equal values at opposite endpoints.

原式0给e^{2x}=1；原函数两端都2+1/2，所以差0。

M1	零点
M1	正确原函数
M1	两端同值
A1	零积分

(b) · 4 marks

1.

左右面积相同，在0至ln2积分原式为(2+.5)−2=.5，再乘2。

M1	分段或对称
M1	正确半面积
M1	乘2
A1	总面积

Question 27

P3-C07-R27 · 6 marks

(a) · 3 marks

$y = \ln(5x + 1) + 2$.

分子正好是真数导数5，ln原函数；初值C2。

M1	对数识别
M1	常数条件
A1	曲线

(b) · 3 marks

$((e^{-2} - 1)/5, 0)$.

ln(5x+1)=−2给5x+1=e^{-2}，所得x大于−1/5合法。

M1	指数反解
M1	域检验
A1	坐标

Question 28

P3-C07-R28 · 6 marks

(a) · 3 marks

$F' = \sec^2 x / \tan x = 1/(\sin x \cos x)$.

ln链式导数，倒数定义化简，域内所有比正。

M1	链式正切导数
M1	三角比化简
A1	目标式

(b) · 3 marks

$\frac{1}{2}\ln 3$.

[ln tanx]=ln1−ln(1/√3)=半ln3。

M1	原函数识别
M1	精确三角代限
A1	值

Question 29

P3-C07-R29 · 6 marks

(a) · 2 marks

$F\prime = (2x-3)e^{2x}$.

乘积法则给$e^{2x} + 2(x-2)e^{2x} = (2x-3)e^{2x}$。

M1	乘积与链式法则
A1	正确整理

(b) Hence, using your answer to (a),

(i) · 2 marks

$2 - e^2$.

直接识别被积函数为$F\prime$，所以积分$F(1) - F(0) = -e^2 - (-2) = 2 - e^2$；不需要分部积分。

M1	由已证明导数识别原函数
A1	精确值

(ii) · 2 marks

$e^2 - 2$.

区间内$2x - 3 < 0$且$e^{2x} > 0$，整条曲线在轴下方，几何面积等于前问有向积分的相反数。

B1	判断曲线符号
A1	正面积

Question 30

P3-C07-R30 · 7 marks

(a) · 3 marks

2.

区间内两项均非负。原函数$\tan x + \sin 2x$，代入端点为$1 + 1 = 2$。

M1	两项标准积分
M1	正确上下限
A1	面积

(b) The line $y = k$ is drawn across the same vertical strip.

(i) · 2 marks

$k = 8/\pi$.

由(a)得$2 - k\pi/4 = 0$，所以$k = 8/\pi$。

M1	用已得积分形成方程
A1	k

(ii) · 2 marks

The areas above and below the line cancel in the signed integral.

零有向积分仅说明正负部分抵消，并不说明每一点的函数差为零；例如$y(0) = 3$而$k = 8/\pi \neq 3$。

B1	有向面积抵消
B1	用给定曲线举例